Vectorized Backpropagation for Neural Network

https://gemini.google.com/app/5d87692d8ac85ca9

User prompt: My Feed-Forward Neural Network has 3 input neurons 2 neurons in hidder layer 1 (ReLU Activation) 3 neurons in hidden layer 2 (ReLU Activation) 2 neurons in hidden layer 3 (ReLU Activation) and 1 neuron in output layer (Sigmoid Activation) My input sample x is a 3×1 matrix. My weights matrix W has a row for each neuron in a given layer. My Bias Matrix is column matrix, where each row represents a neuron in a given layer. My loss function is binary cross-entropy for a single input sample. Can you please vectorize backpropagation?

Response: The vectorized backpropagation derivation for your network architecture uses the following dimensions and layer counts:

- **Layer 0 (Input x):** 3×1
- **Layer 1 ($n^{[1]} = 2$, ReLU):** $W^{[1]} \in \mathbb{R}^{2\times3}$, $b^{[1]} \in \mathbb{R}^{2\times1}$
- **Layer 2 ($n^{[2]} = 3$, ReLU):** $W^{[2]} \in \mathbb{R}^{3\times2}$, $b^{[2]} \in \mathbb{R}^{3\times1}$
- **Layer 3 ($n^{[3]} = 2$, ReLU):** $W^{[3]} \in \mathbb{R}^{2\times3}$, $b^{[3]} \in \mathbb{R}^{2\times1}$
- **Layer 4 ($n^{[4]} = 1$, Sigmoid):** $W^{[4]} \in \mathbb{R}^{1\times2}$, $b^{[4]} \in \mathbb{R}^{1\times1}$

Forward Pass Summary

For layer $l \in \{1, 2, 3, 4\}$ with input $a^{[0]} = x$:

$$z^{[l]} = W^{[l]}a^{[l-1]} + b^{[l]}$$

$$a^{[l]} = g^{[l]}(z^{[l]})$$

The single-sample Binary Cross-Entropy loss is:

$$\mathcal{L}(a^{[4]}, y) = -\left[y\log(a^{[4]}) + (1-y)\log(1-a^{[4]})\right]$$

Step 1: Output Layer Gradient ($\delta^{[4]}$)

Define the error term at layer l as $\delta^{[l]} = \frac{\partial \mathcal{L}}{\partial z^{[l]}}$.

Combining the derivative of the Binary Cross-Entropy loss with respect to $a^{[4]}$ and the derivative of the Sigmoid activation $g'^{[4]}(z^{[4]}) = a^{[4]}(1-a^{[4]})$ gives:

$$\delta^{[4]} = \frac{\partial \mathcal{L}}{\partial a^{[4]}} \odot g'^{[4]}(z^{[4]}) = a^{[4]} - y \quad (1\times1)$$

Gradients for Layer 4 parameters:

$$\frac{\partial \mathcal{L}}{\partial W^{[4]}} = \delta^{[4]}(a^{[3]})^T \quad (1\times1)\times(1\times2) = (1\times2)$$

$$\frac{\partial \mathcal{L}}{\partial b^{[4]}} = \delta^{[4]} \quad (1\times1)$$

Step 2: Recursive Backpropagation Step

For any hidden layer $l \in \{1, 2, 3\}$, the error vector propagates backward according to:

$$\delta^{[l]} = \left((W^{[l+1]})^T\delta^{[l+1]}\right) \odot g'^{[l]}(z^{[l]})$$

Where $\odot$ represents element-wise multiplication (Hadamard product), and $g'^{[l]}(z^{[l]})$ is the derivative of the ReLU function:

$$g'^{[l]}(z^{[l]}) = \begin{cases} 1 & \text{if } z^{[l]} > 0 \\ 0 & \text{otherwise} \end{cases}$$

Parameter gradients at layer l:

$$\frac{\partial \mathcal{L}}{\partial W^{[l]}} = \delta^{[l]}(a^{[l-1]})^T$$
$$\frac{\partial \mathcal{L}}{\partial b^{[l]}} = \delta^{[l]}$$

Step 3: Layer-by-Layer Calculation

Hidden Layer 3 ($n^{[3]} = 2$):

$$\delta^{[3]} = \left((W^{[4]})^T\delta^{[4]}\right) \odot \mathbb{I}(z^{[3]} > 0) \quad ((2\times 1)\times(1\times 1)) \odot (2\times 1) = (2\times 1)$$
$$\frac{\partial \mathcal{L}}{\partial W^{[3]}} = \delta^{[3]}(a^{[2]})^T \quad (2\times 1)\times(1\times 3) = (2\times 3)$$
$$\frac{\partial \mathcal{L}}{\partial b^{[3]}} = \delta^{[3]} \quad (2\times 1)$$

Hidden Layer 2 ($n^{[2]} = 3$):

$$\delta^{[2]} = \left((W^{[3]})^T\delta^{[3]}\right) \odot \mathbb{I}(z^{[2]} > 0) \quad ((3\times 2)\times(2\times 1)) \odot (3\times 1) = (3\times 1)$$
$$\frac{\partial \mathcal{L}}{\partial W^{[2]}} = \delta^{[2]}(a^{[1]})^T \quad (3\times 1)\times(1\times 2) = (3\times 2)$$
$$\frac{\partial \mathcal{L}}{\partial b^{[2]}} = \delta^{[2]} \quad (3\times 1)$$

Hidden Layer 1 ($n^{[1]} = 2$):

$$\delta^{[1]} = \left((W^{[2]})^T\delta^{[2]}\right) \odot \mathbb{I}(z^{[1]} > 0) \quad ((2\times 3)\times(3\times 1)) \odot (2\times 1) = (2\times 1)$$
$$\frac{\partial \mathcal{L}}{\partial W^{[1]}} = \delta^{[1]}(x)^T \quad (2\times 1)\times(1\times 3) = (2\times 3)$$
$$\frac{\partial \mathcal{L}}{\partial b^{[1]}} = \delta^{[1]} \quad (2\times 1)$$

Summary of Gradient Matrix Dimensions

Layer (l)	Weight Matrix ($W^{[l]}$)	Gradient ($\frac{\partial \mathcal{L}}{\partial W^{[l]}}$)	Bias Vector ($b^{[l]}$) & Gradient ($\frac{\partial \mathcal{L}}{\partial b^{[l]}}$)
Layer 1	2×3	2×3	2×1
Layer 2	3×2	3×2	3×1
Layer 3	2×3	2×3	2×1
Layer 4	1×2	1×2	1×1

User prompt: Compute the gradients for input sample, x = [1, 1, 1], which is a column vector. Output Layer Ignore dL/dA. Compute dL/DZ, dZ/dW, dZ/dB, and combine them to compute dL/dW, dL/dB Hidden Layers Compute dZ/dW, dZ/dB, dA/dZ, d(Z of next layer)/dA, then combine them to compute dL/dW, dL/dB.

Response: For input $x = \begin{bmatrix}1\\1\\1\end{bmatrix}$, here is the step-by-step breakdown of the requested derivative terms and their matrix combinations.

Output Layer (Layer 4)

1. Gradient components:

- $\frac{\partial \mathcal{L}}{\partial z^{[4]}}$ **($\delta^{[4]}$):** Combining the Binary Cross-Entropy derivative with the Sigmoid activation derivative yields:

$$\frac{\partial \mathcal{L}}{\partial z^{[4]}} = a^{[4]} - y \quad (1 \times 1)$$

- $\frac{\partial z^{[4]}}{\partial W^{[4]}}$: Since $z^{[4]} = W^{[4]}a^{[3]} + b^{[4]}$, the rate of change with respect to the weights is:

$$\frac{\partial z^{[4]}}{\partial W^{[4]}} = (a^{[3]})^T \quad (1 \times 2)$$

- $\frac{\partial z^{[4]}}{\partial b^{[4]}}$:

$$\frac{\partial z^{[4]}}{\partial b^{[4]}} = 1 \quad (1 \times 1)$$

2. Combining for weight and bias gradients:

- $\frac{\partial \mathcal{L}}{\partial W^{[4]}}$:

$$\frac{\partial \mathcal{L}}{\partial W^{[4]}} = \frac{\partial \mathcal{L}}{\partial z^{[4]}} \frac{\partial z^{[4]}}{\partial W^{[4]}} = (a^{[4]} - y)(a^{[3]})^T \quad (1 \times 1) \times (1 \times 2) = (1 \times 2)$$

- $\frac{\partial \mathcal{L}}{\partial b^{[4]}}$:

$$\frac{\partial \mathcal{L}}{\partial b^{[4]}} = \frac{\partial \mathcal{L}}{\partial z^{[4]}} \frac{\partial z^{[4]}}{\partial b^{[4]}} = a^{[4]} - y \quad (1 \times 1)$$

Hidden Layers (Layers $l \in \{1, 2, 3\}$)

1. Derivative terms:

- $\frac{\partial z^{[l]}}{\partial W^{[l]}}$:

$$\frac{\partial z^{[l]}}{\partial W^{[l]}} = (a^{[l-1]})^T$$

- $\frac{\partial z^{[l]}}{\partial b^{[l]}}$:

$$\frac{\partial z^{[l]}}{\partial b^{[l]}} = I_{n^{[l]}} \quad (\text{Identity matrix of size } n^{[l]})$$

- $\frac{\partial a^{[l]}}{\partial z^{[l]}}$: For ReLU activation:

$$\frac{\partial a^{[l]}}{\partial z^{[l]}} = \text{diag}(\mathbb{I}(z^{[l]} > 0))$$

- $\frac{\partial z^{[l+1]}}{\partial a^{[l]}}$: Since $z^{[l+1]} = W^{[l+1]}a^{[l]} + b^{[l+1]}$:

$$\frac{\partial z^{[l+1]}}{\partial a^{[l]}} = W^{[l+1]}$$

2. Combining for Hidden Layer gradients:

First, find $\delta^{[l]} = \frac{\partial \mathcal{L}}{\partial z^{[l]}}$ by applying the chain rule backward from layer $l + 1$:

$$\delta^{[l]} = \left(\frac{\partial z^{[l+1]}}{\partial a^{[l]}}\right)^T \frac{\partial \mathcal{L}}{\partial z^{[l+1]}} \odot \frac{\partial a^{[l]}}{\partial z^{[l]}} = \left((W^{[l+1]})^T \delta^{[l+1]}\right) \odot \mathbb{I}(z^{[l]} > 0)$$

Then combine $\delta^{[l]}$ with $\frac{\partial z^{[l]}}{\partial W^{[l]}}$ and $\frac{\partial z^{[l]}}{\partial b^{[l]}}$:

$$\frac{\partial \mathcal{L}}{\partial W^{[l]}} = \delta^{[l]}(a^{[l-1]})^T$$

$$\frac{\partial \mathcal{L}}{\partial b^{[l]}} = \delta^{[l]}$$

Explicit Calculations by Layer

Layer 3 ($n^{[3]} = 2$)

- $\delta^{[3]} = \left((W^{[4]})^T(a^{[4]} - y)\right) \odot \mathbb{I}(z^{[3]} > 0) \quad (2 \times 1)$
- $\frac{\partial \mathcal{L}}{\partial W^{[3]}} = \delta^{[3]}(a^{[2]})^T \quad (2 \times 1) \times (1 \times 3) = (2 \times 3)$
- $\frac{\partial \mathcal{L}}{\partial b^{[3]}} = \delta^{[3]} \quad (2 \times 1)$

Layer 2 ($n^{[2]} = 3$)

- $\delta^{[2]} = \left((W^{[3]})^T\delta^{[3]}\right) \odot \mathbb{I}(z^{[2]} > 0) \quad (3 \times 1)$
- $\frac{\partial \mathcal{L}}{\partial W^{[2]}} = \delta^{[2]}(a^{[1]})^T \quad (3 \times 1) \times (1 \times 2) = (3 \times 2)$
- $\frac{\partial \mathcal{L}}{\partial b^{[2]}} = \delta^{[2]} \quad (3 \times 1)$

Layer 1 ($n^{[1]} = 2$, with $x = [1, 1, 1]^T$)

- $\delta^{[1]} = \left((W^{[2]})^T\delta^{[2]}\right) \odot \mathbb{I}(z^{[1]} > 0) \quad (2 \times 1)$
- $\frac{\partial z^{[1]}}{\partial W^{[1]}} = x^T = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial W^{[1]}} = \delta^{[1]} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \delta_1^{[1]} & \delta_1^{[1]} & \delta_1^{[1]} \\ \delta_2^{[1]} & \delta_2^{[1]} & \delta_2^{[1]} \end{bmatrix} \quad (2 \times 3)$
- $\frac{\partial \mathcal{L}}{\partial b^{[1]}} = \delta^{[1]} = \begin{bmatrix} \delta_1^{[1]} \\ \delta_2^{[1]} \end{bmatrix} \quad (2 \times 1)$

User prompt: please, compute assuming all bias and weights is 1

Response: Step 1: Forward Pass

With $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and all weight entries $W_{i,j}^{[l]} = 1$ and bias entries $b_i^{[l]} = 1$:

Layer 1:

- $z^{[1]} = W^{[1]}x + b^{[1]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$
- $a^{[1]} = \text{ReLU}(z^{[1]}) = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$

Layer 2:

- $z^{[2]} = W^{[2]}a^{[1]} + b^{[2]} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 9 \\ 9 \end{bmatrix}$
- $a^{[2]} = \text{ReLU}(z^{[2]}) = \begin{bmatrix} 9 \\ 9 \\ 9 \end{bmatrix}$

Layer 3:

- $z^{[3]} = W^{[3]}a^{[2]} + b^{[3]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 9 \\ 9 \\ 9 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 28 \\ 28 \end{bmatrix}$

- $a^{[3]} = \text{ReLU}(z^{[3]}) = \begin{bmatrix} 28 \\ 28 \end{bmatrix}$

Layer 4 (Output):

- $z^{[4]} = W^{[4]}a^{[3]} + b^{[4]} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 28 \\ 28 \end{bmatrix} + 1 = 57$
- $a^{[4]} = \sigma(57) = \frac{1}{1+e^{-57}} \approx 1.0$

Step 2: Backward Pass (Error Terms $\delta^{[l]}$)

Define $\Delta = a^{[4]} - y = \sigma(57) - y \approx 1 - y$.

Since all $z^{[l]} > 0$ for hidden layers, the derivative of ReLU is 1 for all units:

- **Layer 4:** $\delta^{[4]} = a^{[4]} - y = \Delta$
- **Layer 3:** $\delta^{[3]} = (W^{[4]})^T \delta^{[4]} \odot 1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Delta = \begin{bmatrix} \Delta \\ \Delta \end{bmatrix}$
- **Layer 2:** $\delta^{[2]} = (W^{[3]})^T \delta^{[3]} \odot 1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \Delta \\ \Delta \end{bmatrix} = \begin{bmatrix} 2\Delta \\ 2\Delta \\ 2\Delta \end{bmatrix}$
- **Layer 1:** $\delta^{[1]} = (W^{[2]})^T \delta^{[2]} \odot 1 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2\Delta \\ 2\Delta \\ 2\Delta \end{bmatrix} = \begin{bmatrix} 6\Delta \\ 6\Delta \end{bmatrix}$

Step 3: Numerical Gradients

Layer 4

- $\frac{\partial \mathcal{L}}{\partial W^{[4]}} = \delta^{[4]}(a^{[3]})^T = \Delta \begin{bmatrix} 28 & 28 \end{bmatrix} = \begin{bmatrix} 28\Delta & 28\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[4]}} = \delta^{[4]} = \Delta$

Layer 3

- $\frac{\partial \mathcal{L}}{\partial W^{[3]}} = \delta^{[3]}(a^{[2]})^T = \begin{bmatrix} \Delta \\ \Delta \end{bmatrix} \begin{bmatrix} 9 & 9 & 9 \end{bmatrix} = \begin{bmatrix} 9\Delta & 9\Delta & 9\Delta \\ 9\Delta & 9\Delta & 9\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[3]}} = \delta^{[3]} = \begin{bmatrix} \Delta \\ \Delta \end{bmatrix}$

Layer 2

- $\frac{\partial \mathcal{L}}{\partial W^{[2]}} = \delta^{[2]}(a^{[1]})^T = \begin{bmatrix} 2\Delta \\ 2\Delta \\ 2\Delta \end{bmatrix} \begin{bmatrix} 4 & 4 \end{bmatrix} = \begin{bmatrix} 8\Delta & 8\Delta \\ 8\Delta & 8\Delta \\ 8\Delta & 8\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[2]}} = \delta^{[2]} = \begin{bmatrix} 2\Delta \\ 2\Delta \\ 2\Delta \end{bmatrix}$

Layer 1

- $\frac{\partial \mathcal{L}}{\partial W^{[1]}} = \delta^{[1]}x^T = \begin{bmatrix} 6\Delta \\ 6\Delta \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 6\Delta & 6\Delta & 6\Delta \\ 6\Delta & 6\Delta & 6\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[1]}} = \delta^{[1]} = \begin{bmatrix} 6\Delta \\ 6\Delta \end{bmatrix}$

Concrete Target Evaluated ($y = 0$ vs $y = 1$)

- **If target** $y = 0$: $\Delta = \sigma(57) - 0 \approx 1$
  - $\frac{\partial \mathcal{L}}{\partial W^{[4]}} \approx \begin{bmatrix} 28 & 28 \end{bmatrix}$
  - $\frac{\partial \mathcal{L}}{\partial W^{[3]}} \approx \begin{bmatrix} 9 & 9 & 9 \\ 9 & 9 & 9 \end{bmatrix}$
  - $\frac{\partial \mathcal{L}}{\partial W^{[2]}} \approx \begin{bmatrix} 8 & 8 \\ 8 & 8 \\ 8 & 8 \end{bmatrix}$
  - $\frac{\partial \mathcal{L}}{\partial W^{[1]}} \approx \begin{bmatrix} 6 & 6 & 6 \\ 6 & 6 & 6 \end{bmatrix}$
- **If target** $y = 1$: $\Delta = \sigma(57) - 1 \approx 0$ (due to saturation at $z^{[4]} = 57$), causing all gradients to vanish near 0.

User prompt: compute using w_ij = 0.1 if j =1, 0.2 if j=2, and 0.3 if j=3; and b_i = 0.1 if i=1, 0.2 if i=2, 0.3 if i=3.

Response: Here is the complete calculation using your specific indexed weights W_{ij} and biases b_i.

Matrix Definitions

Based on W_{ij} (row i, column j) and b_i:

- **Layer 1** (2×3 weights, 2×1 bias):

$$W^{[1]} = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix}, \quad b^{[1]} = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$$

- **Layer 2** (3×2 weights, 3×1 bias):

$$W^{[2]} = \begin{bmatrix} 0.1 & 0.2 \\ 0.1 & 0.2 \\ 0.1 & 0.2 \end{bmatrix}, \quad b^{[2]} = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix}$$

- **Layer 3** (2×3 weights, 2×1 bias):

$$W^{[3]} = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix}, \quad b^{[3]} = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$$

- **Layer 4** (1×2 weights, 1×1 bias):

$$W^{[4]} = \begin{bmatrix} 0.1 & 0.2 \end{bmatrix}, \quad b^{[4]} = \begin{bmatrix} 0.1 \end{bmatrix}$$

Step 1: Forward Pass

With input $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$:

Layer 1:

- $z^{[1]} = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.6 + 0.1 \\ 0.6 + 0.2 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.8 \end{bmatrix}$
- $a^{[1]} = \text{ReLU}(z^{[1]}) = \begin{bmatrix} 0.7 \\ 0.8 \end{bmatrix}$

Layer 2:

- $z^{[2]} = \begin{bmatrix} 0.1 & 0.2 \\ 0.1 & 0.2 \\ 0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.8 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} = \begin{bmatrix} 0.23 + 0.1 \\ 0.23 + 0.2 \\ 0.23 + 0.3 \end{bmatrix} = \begin{bmatrix} 0.33 \\ 0.43 \\ 0.53 \end{bmatrix}$
- $a^{[2]} = \text{ReLU}(z^{[2]}) = \begin{bmatrix} 0.33 \\ 0.43 \\ 0.53 \end{bmatrix}$

Layer 3:

- $z^{[3]} = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 0.33 \\ 0.43 \\ 0.53 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.278 + 0.1 \\ 0.278 + 0.2 \end{bmatrix} = \begin{bmatrix} 0.378 \\ 0.478 \end{bmatrix}$
- $a^{[3]} = \text{ReLU}(z^{[3]}) = \begin{bmatrix} 0.378 \\ 0.478 \end{bmatrix}$

Layer 4 (Output):

- $z^{[4]} = \begin{bmatrix} 0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 0.378 \\ 0.478 \end{bmatrix} + 0.1 = 0.1334 + 0.1 = 0.2334$
- $a^{[4]} = \sigma(0.2334) = \frac{1}{1+e^{-0.2334}} \approx 0.55806$

Step 2: Backward Pass (Error Terms $\delta^{[l]}$)

Let $\Delta = a^{[4]} - y \approx 0.55806 - y$.

Since all $z^{[l]} > 0$, the derivative of ReLU is 1 for all units:

- **Layer 4:**

$$\delta^{[4]} = \Delta$$

- **Layer 3:**

$$\delta^{[3]} = (W^{[4]})^T \delta^{[4]} \odot 1 = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} \Delta = \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix}$$

- **Layer 2:**

$$\delta^{[2]} = (W^{[3]})^T \delta^{[3]} \odot 1 = \begin{bmatrix} 0.1 & 0.1 \\ 0.2 & 0.2 \\ 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix} = \begin{bmatrix} 0.03\Delta \\ 0.06\Delta \\ 0.09\Delta \end{bmatrix}$$

- **Layer 1:**

$$\delta^{[1]} = (W^{[2]})^T \delta^{[2]} \odot 1 = \begin{bmatrix} 0.1 & 0.1 & 0.1 \\ 0.2 & 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} 0.03\Delta \\ 0.06\Delta \\ 0.09\Delta \end{bmatrix} = \begin{bmatrix} 0.018\Delta \\ 0.036\Delta \end{bmatrix}$$

Step 3: Gradient Matrices

Layer 4

- $\frac{\partial \mathcal{L}}{\partial W^{[4]}} = \delta^{[4]} (a^{[3]})^T = \Delta \begin{bmatrix} 0.378 & 0.478 \end{bmatrix} = \begin{bmatrix} 0.378\Delta & 0.478\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[4]}} = \delta^{[4]} = \Delta$

Layer 3

- $\frac{\partial \mathcal{L}}{\partial W^{[3]}} = \delta^{[3]} (a^{[2]})^T = \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix} \begin{bmatrix} 0.33 & 0.43 & 0.53 \end{bmatrix} = \begin{bmatrix} 0.033\Delta & 0.043\Delta & 0.053\Delta \\ 0.066\Delta & 0.086\Delta & 0.106\Delta \end{bmatrix}$

- $\frac{\partial \mathcal{L}}{\partial b^{[3]}} = \delta^{[3]} = \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix}$

Layer 2

- $\frac{\partial \mathcal{L}}{\partial W^{[2]}} = \delta^{[2]} (a^{[1]})^T = \begin{bmatrix} 0.03\Delta \\ 0.06\Delta \\ 0.09\Delta \end{bmatrix} \begin{bmatrix} 0.7 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.021\Delta & 0.024\Delta \\ 0.042\Delta & 0.048\Delta \\ 0.063\Delta & 0.072\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[2]}} = \delta^{[2]} = \begin{bmatrix} 0.03\Delta \\ 0.06\Delta \\ 0.09\Delta \end{bmatrix}$

Layer 1

- $\frac{\partial \mathcal{L}}{\partial W^{[1]}} = \delta^{[1]} x^T = \begin{bmatrix} 0.018\Delta \\ 0.036\Delta \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.018\Delta & 0.018\Delta & 0.018\Delta \\ 0.036\Delta & 0.036\Delta & 0.036\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[1]}} = \delta^{[1]} = \begin{bmatrix} 0.018\Delta \\ 0.036\Delta \end{bmatrix}$

Numerical Output for Targets $y = 0$ and $y = 1$

Parameter Gradient	Formula with Δ	Value if $y = 0$ ($\Delta = 0.55806$)	Value if $y = 1$ ($\Delta = -0.44194$)
$\frac{\partial \mathcal{L}}{\partial W^{[4]}}$	$\begin{bmatrix} 0.378\Delta & 0.478\Delta \end{bmatrix}$	$\begin{bmatrix} 0.21095 & 0.26675 \end{bmatrix}$	$\begin{bmatrix} -0.16705 & -0.21125 \end{bmatrix}$
$\frac{\partial \mathcal{L}}{\partial W^{[3]}}$	$\begin{bmatrix} 0.033\Delta & 0.043\Delta & 0.053\Delta \\ 0.066\Delta & 0.086\Delta & 0.106\Delta \end{bmatrix}$	$\begin{bmatrix} 0.01842 & 0.02400 & 0.02958 \\ 0.03683 & 0.04799 & 0.05915 \end{bmatrix}$	$\begin{bmatrix} -0.01458 & -0.01900 & -0.02342 \\ -0.02917 & -0.03801 & -0.04685 \end{bmatrix}$
$\frac{\partial \mathcal{L}}{\partial W^{[2]}}$	$\begin{bmatrix} 0.021\Delta & 0.024\Delta \\ 0.042\Delta & 0.048\Delta \\ 0.063\Delta & 0.072\Delta \end{bmatrix}$	$\begin{bmatrix} 0.01172 & 0.01339 \\ 0.02344 & 0.02679 \\ 0.03516 & 0.04018 \end{bmatrix}$	$\begin{bmatrix} -0.00928 & -0.01061 \\ -0.01856 & -0.02121 \\ -0.02784 & -0.03182 \end{bmatrix}$
$\frac{\partial \mathcal{L}}{\partial W^{[1]}}$	$\begin{bmatrix} 0.018\Delta & 0.018\Delta & 0.018\Delta \\ 0.036\Delta & 0.036\Delta & 0.036\Delta \end{bmatrix}$	$\begin{bmatrix} 0.01004 & 0.01004 & 0.01004 \\ 0.02009 & 0.02009 & 0.02009 \end{bmatrix}$	$\begin{bmatrix} -0.00795 & -0.00795 & -0.00795 \\ -0.01591 & -0.01591 & -0.01591 \end{bmatrix}$

User prompt: be explicit on following during graidients calculation Output Layer Ignore dL/dA. Compute dL/DZ, dZ/dW, dZ/dB, and combine them to compute dL/dW, dL/dB Hidden Layers Compute dZ/dW, dZ/dB, dA/dZ, d(Z of next layer)/dA, then combine them to compute dL/dW, dL/dB.

Response: Here is the explicit step-by-step breakdown using your exact numerical values from the forward pass ($a^{[4]} \approx 0.55806$, $\Delta = a^{[4]} - y$).

Output Layer (Layer 4)

1. Individual Partial Derivatives:

- $\frac{\partial \mathcal{L}}{\partial z^{[4]}} = \Delta$
- $\frac{\partial z^{[4]}}{\partial W^{[4]}} = (a^{[3]})^T = \begin{bmatrix} 0.378 & 0.478 \end{bmatrix}$
- $\frac{\partial z^{[4]}}{\partial b^{[4]}} = 1$

2. Combining for Final Gradients:

- $\frac{\partial \mathcal{L}}{\partial W^{[4]}} = \frac{\partial \mathcal{L}}{\partial z^{[4]}} \cdot \frac{\partial z^{[4]}}{\partial W^{[4]}} = \Delta \cdot \begin{bmatrix} 0.378 & 0.478 \end{bmatrix} = \begin{bmatrix} 0.378\Delta & 0.478\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[4]}} = \frac{\partial \mathcal{L}}{\partial z^{[4]}} \cdot \frac{\partial z^{[4]}}{\partial b^{[4]}} = \Delta \cdot 1 = \Delta$

Hidden Layer 3

1. Individual Partial Derivatives:

- $\frac{\partial z^{[3]}}{\partial W^{[3]}} = (a^{[2]})^T = \begin{bmatrix} 0.33 & 0.43 & 0.53 \end{bmatrix}$
- $\frac{\partial z^{[3]}}{\partial b^{[3]}} = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\frac{\partial a^{[3]}}{\partial z^{[3]}} = \begin{bmatrix} \mathbb{I}(z_1^{[3]} > 0) & 0 \\ 0 & \mathbb{I}(z_2^{[3]} > 0) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\frac{\partial z^{[4]}}{\partial a^{[3]}} = W^{[4]} = \begin{bmatrix} 0.1 & 0.2 \end{bmatrix}$

2. Combining for Final Gradients:

- $\delta^{[3]} = \frac{\partial \mathcal{L}}{\partial z^{[3]}} = \left(\frac{\partial z^{[4]}}{\partial a^{[3]}}\right)^T \frac{\partial \mathcal{L}}{\partial z^{[4]}} \odot \operatorname{diag}\left(\frac{\partial a^{[3]}}{\partial z^{[3]}}\right) = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} \Delta \odot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial W^{[3]}} = \delta^{[3]} \cdot \frac{\partial z^{[3]}}{\partial W^{[3]}} = \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix} \begin{bmatrix} 0.33 & 0.43 & 0.53 \end{bmatrix} = \begin{bmatrix} 0.033\Delta & 0.043\Delta & 0.053\Delta \\ 0.066\Delta & 0.086\Delta & 0.106\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[3]}} = \delta^{[3]} \cdot \frac{\partial z^{[3]}}{\partial b^{[3]}} = \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix}$

Hidden Layer 2

1. Individual Partial Derivatives:

- $\frac{\partial z^{[2]}}{\partial W^{[2]}} = (a^{[1]})^T = \begin{bmatrix} 0.7 & 0.8 \end{bmatrix}$
- $\frac{\partial z^{[2]}}{\partial b^{[2]}} = I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $\frac{\partial a^{[2]}}{\partial z^{[2]}} = \operatorname{diag}(\mathbb{I}(z^{[2]} > 0)) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $\frac{\partial z^{[3]}}{\partial a^{[2]}} = W^{[3]} = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix}$

2. Combining for Final Gradients:

- $\delta^{[2]} = \frac{\partial \mathcal{L}}{\partial z^{[2]}} = \left(\frac{\partial z^{[3]}}{\partial a^{[2]}}\right)^T \delta^{[3]} \odot \operatorname{diag}\left(\frac{\partial a^{[2]}}{\partial z^{[2]}}\right) = \begin{bmatrix} 0.1 & 0.1 \\ 0.2 & 0.2 \\ 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix} \odot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.03\Delta \\ 0.06\Delta \\ 0.09\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial W^{[2]}} = \delta^{[2]} \cdot \frac{\partial z^{[2]}}{\partial W^{[2]}} = \begin{bmatrix} 0.03\Delta \\ 0.06\Delta \\ 0.09\Delta \end{bmatrix} \begin{bmatrix} 0.7 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.021\Delta & 0.024\Delta \\ 0.042\Delta & 0.048\Delta \\ 0.063\Delta & 0.072\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[2]}} = \delta^{[2]} \cdot \frac{\partial z^{[2]}}{\partial b^{[2]}} = \begin{bmatrix} 0.03\Delta \\ 0.06\Delta \\ 0.09\Delta \end{bmatrix}$

Hidden Layer 1

1. Individual Partial Derivatives:

- $\frac{\partial z^{[1]}}{\partial W^{[1]}} = x^T = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- $\frac{\partial z^{[1]}}{\partial b^{[1]}} = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

- $\frac{\partial a^{[1]}}{\partial z^{[1]}} = \text{diag}(\mathbb{I}(z^{[1]} > 0)) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\frac{\partial z^{[2]}}{\partial a^{[1]}} = W^{[2]} = \begin{bmatrix} 0.1 & 0.2 \\ 0.1 & 0.2 \\ 0.1 & 0.2 \end{bmatrix}$

2. Combining for Final Gradients:

- $\delta^{[1]} = \frac{\partial \mathcal{L}}{\partial z^{[1]}} = \left(\frac{\partial z^{[2]}}{\partial a^{[1]}}\right)^T \delta^{[2]} \odot \text{diag}\left(\frac{\partial a^{[1]}}{\partial z^{[1]}}\right) = \begin{bmatrix} 0.1 & 0.1 & 0.1 \\ 0.2 & 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} 0.03\Delta \\ 0.06\Delta \\ 0.09\Delta \end{bmatrix} \odot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.018\Delta \\ 0.036\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial W^{[1]}} = \delta^{[1]} \cdot \frac{\partial z^{[1]}}{\partial W^{[1]}} = \begin{bmatrix} 0.018\Delta \\ 0.036\Delta \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.018\Delta & 0.018\Delta & 0.018\Delta \\ 0.036\Delta & 0.036\Delta & 0.036\Delta \end{bmatrix}$
- $\frac{\partial \mathcal{L}}{\partial b^{[1]}} = \delta^{[1]} \cdot \frac{\partial z^{[1]}}{\partial b^{[1]}} = \begin{bmatrix} 0.018\Delta \\ 0.036\Delta \end{bmatrix}$

User prompt: Why identity matrix?

Response: The term $\frac{\partial z^{[l]}}{\partial b^{[l]}}$ is an **Identity Matrix (I)** because each bias term $b_i^{[l]}$ affects **only** its corresponding linear output $z_i^{[l]}$, and no other neuron's output.

1. Element-by-Element Look

For layer l, the linear equation for neuron i is:

$$z_i^{[l]} = \sum_j W_{ij}^{[l]} a_j^{[l-1]} + b_i^{[l]}$$

Taking the partial derivative of $z_i^{[l]}$ with respect to each bias $b_k^{[l]}$:

- **When $i = k$:** $\frac{\partial z_i^{[l]}}{\partial b_i^{[l]}} = 1$ (changing b_i directly changes z_i by the exact same amount)
- **When $i \neq k$:** $\frac{\partial z_i^{[l]}}{\partial b_k^{[l]}} = 0$ (b_k has zero impact on z_i)

2. The Jacobian Matrix

Arranging all these partial derivatives into a matrix (a Jacobian):

$$\frac{\partial z^{[l]}}{\partial b^{[l]}} = \begin{bmatrix} \frac{\partial z_1}{\partial b_1} & \frac{\partial z_1}{\partial b_2} & \cdots \\ \frac{\partial z_2}{\partial b_1} & \frac{\partial z_2}{\partial b_2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

3. Why It Simplifies

When applying the chain rule to find the loss gradient with respect to bias:

$$\frac{\partial \mathcal{L}}{\partial b^{[l]}} = \delta^{[l]} \cdot \frac{\partial z^{[l]}}{\partial b^{[l]}} = \delta^{[l]} \cdot I = \delta^{[l]}$$

Because multiplying any matrix by I leaves it unchanged, the bias gradient simply reduces directly to the error term $\delta^{[l]}$.

User prompt: $\frac{\partial \mathcal{L}}{\partial b^{[3]}} = \delta^{[3]} \cdot \frac{\partial z^{[3]}}{\partial b^{[3]}} = \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix}$ How do you multiply a 2 by 1 matrix and a 2 by 2 matrix?

Response: To multiply an error vector $\delta^{[3]}$ (2×1) with the Jacobian matrix $\frac{\partial z^{[3]}}{\partial b^{[3]}}$ (2×2), you cannot perform a standard matrix multiplication directly in that order because the inner dimensions don't match: $(2 \times 1) \cdot (2 \times 2)$ **is mathematically undefined**.

Here is how that step is actually calculated correctly under multivariate chain rule principles:

1. Vectorized Chain Rule Transpose Rule

When applying the chain rule to column vectors, the Jacobian of the downstream variable is placed on the **left**, transposed:

$$\frac{\partial \mathcal{L}}{\partial b^{[3]}} = \left(\frac{\partial z^{[3]}}{\partial b^{[3]}}\right)^T \delta^{[3]}$$

Because $\frac{\partial z^{[3]}}{\partial b^{[3]}} = I_2$ (the 2×2 identity matrix):

$$(I_2)^T \delta^{[3]} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix} = \begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix}$$

Here, the dimensions are $(2 \times 2) \times (2 \times 1) = (2 \times 1)$, which is valid.

2. Component-by-Component Summation

Using the scalar multivariable chain rule for each bias element b_i:

$$\frac{\partial \mathcal{L}}{\partial b_i^{[3]}} = \sum_{k=1}^{2} \frac{\partial \mathcal{L}}{\partial z_k^{[3]}} \frac{\partial z_k^{[3]}}{\partial b_i^{[3]}}$$

- **For $b_1^{[3]}$:**

$$\frac{\partial \mathcal{L}}{\partial b_1^{[3]}} = \delta_1^{[3]} \cdot \frac{\partial z_1^{[3]}}{\partial b_1^{[3]}} + \delta_2^{[3]} \cdot \frac{\partial z_2^{[3]}}{\partial b_1^{[3]}} = (0.1\Delta)(1) + (0.2\Delta)(0) = 0.1\Delta$$

- **For $b_2^{[3]}$:**

$$\frac{\partial \mathcal{L}}{\partial b_2^{[3]}} = \delta_1^{[3]} \cdot \frac{\partial z_1^{[3]}}{\partial b_2^{[3]}} + \delta_2^{[3]} \cdot \frac{\partial z_2^{[3]}}{\partial b_2^{[3]}} = (0.1\Delta)(0) + (0.2\Delta)(1) = 0.2\Delta$$

Stacking these two results into a column vector yields $\begin{bmatrix} 0.1\Delta \\ 0.2\Delta \end{bmatrix}$.

Summary

In shorthand notation, writing $\delta^{[3]} \cdot I$ represents element-wise scaling or post-multiplication, but formally in matrix calculus it is evaluated as:

$$\frac{\partial \mathcal{L}}{\partial b^{[3]}} = \left(\frac{\partial z^{[3]}}{\partial b^{[3]}}\right)^T \delta^{[3]} = I \cdot \delta^{[3]} = \delta^{[3]}$$

User prompt: How is $\frac{\partial z^{[3]}}{\partial W^{[3]}} = (a^{[2]})^T = \begin{bmatrix} 0.33 & 0.43 & 0.53 \end{bmatrix}$?

Response: The dimension mismatch you're noticing comes from an abbreviation in shorthand notation.

If we write out the full Jacobian matrix for $\frac{\partial z^{[3]}}{\partial W^{[3]}}$, it is actually a **3D tensor** (a derivative of a 2D vector with respect to a 2×3 matrix). To avoid 3D tensors in practice, standard neural network derivations treat $(a^{[2]})^T$ as the factor that aligns dimensions correctly when computing the gradient matrix.

1. Element-by-Element Derivative

The linear activation for layer 3 is:

$$\begin{bmatrix} z_1^{[3]} \\ z_2^{[3]} \end{bmatrix} = \begin{bmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \end{bmatrix} \begin{bmatrix} a_1^{[2]} \\ a_2^{[2]} \\ a_3^{[2]} \end{bmatrix} + \begin{bmatrix} b_1^{[3]} \\ b_2^{[3]} \end{bmatrix}$$

Written as individual scalar equations:

$$z_1^{[3]} = W_{11}a_1^{[2]} + W_{12}a_2^{[2]} + W_{13}a_3^{[2]} + b_1^{[3]}$$

$$z_2^{[3]} = W_{21}a_1^{[2]} + W_{22}a_2^{[2]} + W_{23}a_3^{[2]} + b_2^{[3]}$$

Taking the partial derivative of $z_i^{[3]}$ with respect to weight W_{ij}:

$$\frac{\partial z_i^{[3]}}{\partial W_{ij}} = a_j^{[2]}$$

Notice that $z_i^{[3]}$ depends **only** on the j-th entry of $a^{[2]}$ ($a_j^{[2]}$), and the derivative is independent of row index i. Thus, the row vector of inputs $(a^{[2]})^T = \begin{bmatrix} a_1^{[2]} & a_2^{[2]} & a_3^{[2]} \end{bmatrix}$ holds all the partial derivatives $\frac{\partial z_i^{[3]}}{\partial W_{ij}}$.

2. Matching Matrix Dimensions

We know from multivariate calculus that the weight gradient $\frac{\partial \mathcal{L}}{\partial W^{[3]}}$ must match the shape of $W^{[3]}$, which is 2×3:

$$\frac{\partial \mathcal{L}}{\partial W^{[3]}} = \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial W_{11}} & \frac{\partial \mathcal{L}}{\partial W_{12}} & \frac{\partial \mathcal{L}}{\partial W_{13}} \\ \frac{\partial \mathcal{L}}{\partial W_{21}} & \frac{\partial \mathcal{L}}{\partial W_{22}} & \frac{\partial \mathcal{L}}{\partial W_{23}} \end{bmatrix}$$

Using the scalar chain rule for any single weight W_{ij}:

$$\frac{\partial \mathcal{L}}{\partial W_{ij}^{[3]}} = \frac{\partial \mathcal{L}}{\partial z_i^{[3]}} \cdot \frac{\partial z_i^{[3]}}{\partial W_{ij}^{[3]}} = \delta_i^{[3]} \cdot a_j^{[2]}$$

To compute all $\delta_i^{[3]} \cdot a_j^{[2]}$ terms at once using matrix multiplication:

- $\delta^{[3]}$ is 2×1: $\begin{bmatrix} \delta_1^{[3]} \\ \delta_2^{[3]} \end{bmatrix}$
- $(a^{[2]})^T$ is 1×3: $\begin{bmatrix} a_1^{[2]} & a_2^{[2]} & a_3^{[2]} \end{bmatrix}$

Multiplying an outer product:

$$\frac{\partial \mathcal{L}}{\partial W^{[3]}} = \delta^{[3]}(a^{[2]})^T = \begin{bmatrix} \delta_1^{[3]} \\ \delta_2^{[3]} \end{bmatrix} \begin{bmatrix} a_1^{[2]} & a_2^{[2]} & a_3^{[2]} \end{bmatrix} = \begin{bmatrix} \delta_1^{[3]}a_1^{[2]} & \delta_1^{[3]}a_2^{[2]} & \delta_1^{[3]}a_3^{[2]} \\ \delta_2^{[3]}a_1^{[2]} & \delta_2^{[3]}a_2^{[2]} & \delta_2^{[3]}a_3^{[2]} \end{bmatrix}$$

This produces the required 2×3 gradient matrix.

Summary

In neural network backpropagation literature, writing $\frac{\partial z^{[3]}}{\partial W^{[3]}} = (a^{[2]})^T$ is shorthand for saying: *"The vector of partial derivatives with respect to the weight matrix columns is $(a^{[2]})^T$, which must be post-multiplied with the error vector $\delta^{[3]}$ via outer product."*
